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Adaptive Observers for Calibrating Immune Digital Twins in Viral Infections

Angel Rodriguez-Liñan[1], Juan E. Sereno[2], Griselda Quiroz-Compean[1], and E.A. Hernandez-Vargas[2]

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***Abstract*— Software duplicates, "Digital twins", are promising tools to merge known immunological mechanisms with real-time patient-specific clinical data to develop predictive computer simulations of viral infection and immune response. During respiratory infections, host immune cells and viral dynamics are critical clinical markers that could help practitioners decide how to treat an infected patient. This paper describes the problem of estimating immune responses and model parameters, which is fundamental for developing immune digital twins. We derived observability and identifiability properties that were the basis of the proposed observation scheme. Different simulation scenarios show which parameters can be estimated during a respiratory infection.**

I. INTRODUCTION

Creating a software replica of all actionable knowledge of the immune system, an immune digital twin, has the potential to provide knowledge into the many unknowns of biology, human health, and disease [1].

Immune digital twins are central to monitor the progression of the disease and make informed choices about therapy. It is essential to monitor the pathogen load and host responses during respiratory infections using minimally invasive techniques [2].

While surrogate markers employing nonlinear observers have garnered considerable attention in biomedical applications [3]–[5], the use of adaptive nonlinear observers for building medical digital twins in viral infections remains unexplored. Integrating estimation techniques may aid in the development of digital twins for disease prediction, as well as help practitioners develop new therapies (Figure 1). Developing immune digital twins from biological and clinical data would enable personalized medicine [6], [7].

Continuously updating digital twins with experimental data is essential for generating reliable and clinically relevant predictions. However, many existing mathematical models are only parameterized once using a fixed dataset, which severely limits their long-term accuracy. The absence of computational frameworks for real-time or iterative re-parameterization creates a gap between static models and true digital twins.

The outline is as follows. The mathematical model to describe in-host SARS-CoV-2 and the respective analysis for observability and identifiability are shown in Section II.

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[1]Universidad Autónoma de Nuevo León, Facultad de Ingeniería Mecánica y Eléctrica, Ave. Universidad S/N, Cd. Universitaria, San Nicolás de los Garza, N.L., CP 66455 México.

[2]Department of Mathematics, University of Idaho, Moscow, Idaho, 83844–1103, USA. esteban@systemsmedicine.de

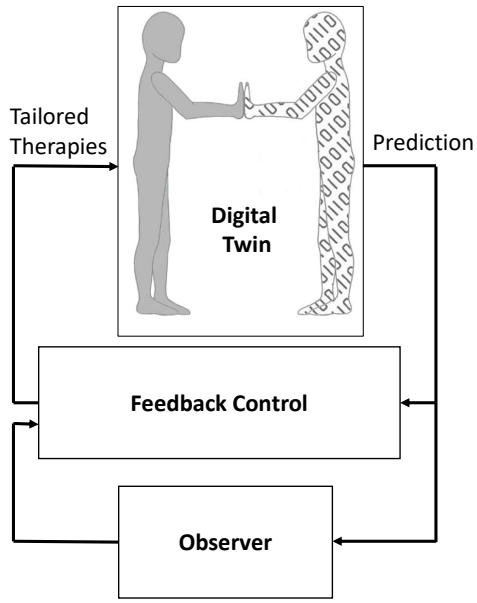

Fig. 1. Control engineering tools for developing immune digital twins. Tailored therapies based on feedback control require information (from the patient and his digital twin) to make predictions, and estimation of state variables and parameters by observers.

After, the design of a state-parameter observer is presented in Section III. Then, the proposed estimation scheme is evaluated by using different simulation scenarios in Section IV. Finally, some conclusions are remarked in Section V.

II. IN-HOST VIRAL DYNAMICAL PROPERTIES

We consider the mathematical model in [8], which is based on two differential equations, to represent the virus (V) and CD8+ T cells (T) dynamics. The model consists of the following equations:

$$\frac{dV}{dt} = pV\left(1-\frac{V}{K}\right) - c_T VT - cV, \quad (1)$$

$$\frac{dT}{dt} = S_T + rT\left(\frac{V^m}{V^m + k_T^m}\right) - \delta_T T. \quad (2)$$

The maximum carrying capacity of the virus is limited by the number of infected cells, which is represented by K. The virus replication rate is p, the normal viral clearance rate is c, and the immune response is represented by $c_T T$. The CD8+ T cells homeostatic replenishment rate is $S_T = \delta_T T_0$. The proliferation rate of CD8 + T cells is represented by r, and δ_T is the half-life of CD8+ T cells. T_0 is the initial number

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of CD8+ T cells, k_T is the half saturation constant for T cells proliferation, and m is a dimensionless coefficient.

A. Observability Analysis

To address the problem of estimating unmeasurable states and parameters of the model described by the system (1)-(2), we analyze the structural observability of the model. This property is called structural since it depends on the information provided by the mathematical model and is based on the assumption that the system's output is a measurable variable. Hence, observability can ensure that it is possible to determine the current state of a system from current and future measurements [9].

To analyze the observability of the system (1)-(2), we first discuss some preliminary concepts of nonlinear systems represented in state space:

$$\dot{X} = f(X), \tag{3}$$
$$Y = h(X), \tag{4}$$

where the state is defined as $X \in D \subset \Re^n$, the output of the system is represented by $Y \in \Re^l$, and $f(\cdot)$, $h(\cdot)$ are smooth functions. The observability is now analyzed from the differential geometric approach defined by Nijmeijer and van der Schaft [10].

Definition 1: Consider the nonlinear system defined in (3)-(4). The observation space $\mathcal{O}$ of this system is the linear space generated by the function $h(\cdot)$ and its Lie derivatives (see Definition 3.29 in [10]).

Representing the model in the general form (3)-(4), $X = [X_1 \ X_2]^T$ with $X_1 := V$ and $X_2 := T$, where the viral load in the host V is the measurable variable, and the output function

$$h(X) = X_1. \tag{5}$$

The observation space of the system (3)-(4) is given by $\mathcal{O} = [h(X) \ L_f h(X)]^T$, where $L_f h(X) = \frac{\partial h(X)}{\partial X} f(X)$. From equation (5), we have $\frac{\partial h(X)}{\partial X} = [1 \ 0]^T$ and $f(X)$ is defined by the right-side of equations (1)-(2). Thus, the observation space becomes:

$$\mathcal{O} = \begin{bmatrix} X_1 \\ pX_1\left(1 - \frac{X_1}{K}\right) - c_T X_1 X_2 - cX_1 \end{bmatrix}. \tag{6}$$

The structural observability proposed in [10] can be seen as a nonlinear generalization of the Kalman rank condition of observability for linear systems using the concept of invariant distributions (Definition 2.37 in [10]). Therefore, the aim here is to analyze if the system (3)-(4) is locally observable at a point $X^0 \in D$.

Definition 2: The observation space $\mathcal{O}$ in (6) defines the observability codistribution $d\mathcal{O}$ setting by

$$d\mathcal{O}(X) = \text{span}(dH(X) \mid H \in \mathcal{O}), \quad X \in \Re^2, \tag{7}$$

where $H_i = \mathcal{O}_i$ and $dH_i(X) = \frac{\partial \mathcal{O}_i}{\partial X}$, for $i = 1, 2$.

From Definition 2 and Theorem 3.32 in [10], we propose the following result.

Theorem 1: The system (3)-(4) is locally observable at $X^0 \in D$, with $X^0 \neq 0$.

Proof: Consider the observation space defined in (6) for a point $X^0 = [X_1^0 \ X_2^0]^T \in D \subset \Re^2$. The calculation of $dH(X^0) = \frac{\partial \mathcal{O}}{\partial X^0}$ results in

$$dH(X_0) = \begin{bmatrix} 1 & 0 \\ p - 2\left(\frac{p}{K}\right)X_1^0 - c_T X_2^0 - c & -c_T X_1^0 \end{bmatrix}.$$

The rows of the above matrix are linearly independent because the rank of the matrix is equal to 2 with $X_1^0 \neq 0$. Then $dim \ d\mathcal{O}(X^0) = 2$ and the system (3)-(4) is locally observable at $X^0 \neq 0$. ■

B. Identifiability Analysis

The algebraic identifiability of the model (1)-(2) is analyzed using the approach reported in [11]. To this end, the model is represented in the following general form:

$$\dot{X} = f(X, u, \Theta), \tag{8}$$
$$Y = h(X), \tag{9}$$

where the state is defined as $X \in \Re^n$, $\Theta \in \Re^v$ is the constant parameter vector, the input of the system $u \in \Re^w$, the output of the system $Y \in \Re^l$, $f(\cdot)$ and $h(\cdot)$ are two smooth functions. According to [11], the system (8)-(9) is algebraically identifiable if there exists a meromorphic function $\Phi(\Theta, u, \dot{u}, ..., u^{(k)}, Y, \dot{Y}, ..., Y^{(k)}) = 0$ such that $\det\left(\frac{\partial \Phi}{\partial \Theta}\right) \neq 0$, where k is a nonnegative integer. In the case of the model (1)-(2) and its representation (8), the input signal is $u \equiv 0$. Therefore, the corresponding function Φ is defined as follows:

$$\Phi = \Phi(\Theta, Y, \dot{Y}, ..., Y^{(k)}), \quad \Phi \in \Re^v. \tag{10}$$

Consider $\Phi : \Re^{v+l} \to \Re^v$ as a function of the parameter vector $\Theta \in \Re^v$, the output $Y \in \Re^l$, and its corresponding time derivatives (from 1st to kth order) as given by (10). Now, assuming that Φ is partially differentiable with respect to Θ, the system (8)-(9) is said to be locally identifiable at a point Θ^0 if there exists a $(k+2)$-upla $(\Theta^0, Y, \dot{Y}, ..., Y^{(k)}) \in \Re^{v+l}$ such that the following conditions hold:

(i) $\Phi(\Theta^0, Y^0, \dot{Y}^0, ..., Y^{0(k)}) = 0$.
(ii) $\det\left(\frac{\partial \Phi}{\partial \Theta^0}\right) \neq 0$.

From equation (10) and Theorem 3.9 in [11], we propose the following result.

Theorem 2: The system (8)-(9) is locally identifiable at $\Theta^0 \in \Re^v$, with $\Theta^0 \neq 0$.

Proof: From the methodology proposed by Xia and Moog [12], let us consider the function Φ as

$$\Phi = \left[\phi \ \phi^{(1)} \ \phi^{(2)} \ \cdots \ \phi^{(v-1)}\right]^T, \tag{11}$$

where $\phi = L_f^n h(X) - Y^{(n)}$. Due to $n = 2$ in the system (8)-(9) and $Y = h(X) = X_1$, then the identification function that satisfies condition (i) becomes

$$\begin{aligned} \phi = & -\left(X_1^{-1}\dot{X}_1 + r^0 G(X_1) - \delta_T^0\right)\left(\alpha X_1 - \sigma X_1^2 - \dot{X}_1\right) \\ & + \alpha \dot{X}_1 - 2\sigma X_1 \dot{X}_1 - \gamma X_1 - \ddot{X}_1, \end{aligned}$$

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where $\alpha = (p^0 - c^0)$, $\sigma = p^0/K^0$, $\gamma = c_T^0 S_T^0$, r^0, and δ_T^0 are functions of unknown parameters of the model (1)-(2), and $G(X_1) = \frac{X_1^m}{X_1^m + k_T^m}$ is a function of X_1, with known parameters m and k_T. On the basis of the above, we have that $\Theta^0 = \left[\alpha\ \sigma\ \gamma\ r^0\ \delta_T^0\right]^T$, then the dimension of the parameter vector Θ^0 becomes $v = 5$. Performing the corresponding derivatives of the identification function ϕ and computing the matrix $\frac{\partial \Phi}{\partial \Theta^0}$, it can be seen that condition (ii) is satisfied,

$$\texttt{rank}\left(\frac{\partial \Phi}{\partial \Theta^0}\right) = 5. \tag{12}$$

Therefore, Θ^0 is locally identifiable. ■

The two above results regarding the structural observability and algebraic identifiability of the model (1)-(2) allow us to address the problem of state and parameter estimation. In particular, due to the variability between patients in the immune system, we propose an adaptive scheme for the estimation of the state variable T and the model parameters $(p - c)$, p/K, $c_T S_T$, r, and δ_T.

III. Adaptive observer for the Immune Response

Consider the model defined by equations (1)-(2). The following change of coordinates

$$\begin{aligned} x_1 &= V/\lambda, \\ x_2 &= \left(pV\left(1 - \tfrac{V}{K}\right) - c_T VT - cV\right)/\lambda, \end{aligned} \tag{13}$$

transforms the model (1)-(2) into

$$\begin{aligned} \dot{x}_1 &= x_2, \\ \dot{x}_2 &= \tfrac{x_2^2}{x_1} + (\alpha\delta_T - \gamma)\, x_1 - \lambda\delta_T\sigma x_1^2 \\ &\quad -r\alpha x_1 G(\lambda x_1) + \lambda r\sigma x_1^2 G(\lambda x_1) \\ &\quad -\delta_T x_2 - \lambda\sigma x_1 x_2 + r x_2 G(\lambda x_1), \end{aligned} \tag{14}$$

where $\lambda \in \Re$ is a known constant scaling factor arbitrarily chosen for convenience of numerical computation. Under the assumption that $y_1 := V/\lambda$ and $y_2 := \dot{V}/\lambda$ are measurable signals for all time, the system (14) can be rewritten in the state affine form with unknown parameters (as in [13])

$$\begin{aligned} \dot{x} &= A(y,u)x + \beta(y,u) + \varphi(y,u)\theta \\ y &= Cx, \end{aligned} \tag{15}$$

where $x \in \Re^n$, $y \in \Re^d$, $u \in \Re^w$ and $\theta \in \Re^q$ are the state vector, measurable output vector, input vector and uncertain constant parameter vector, respectively. Since $x = [x_1\ x_2]^T$ and $y = [y_1\ y_2]^T$, then $d = n = 2$ and $C = I_{2\times 2}$ is the output matrix. Matrices $A(y,u) \in \Re^{n\times n}$, $\beta(y,u) \in \Re^n$ and $\varphi(y,u) \in \Re^{n\times q}$ are uniformly bounded continuous functions that depend only on y, since $u \equiv 0$ for both the model (1)-(2) and its transformed form (14), given by

$$A = \begin{bmatrix} 0 & 1 \\ 0 & y_2/y_1 \end{bmatrix}, \qquad \beta = \begin{bmatrix} 0 \\ 0 \end{bmatrix},$$

$$\varphi = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ y_1 & \varphi_{2,2} & \varphi_{2,3} & \varphi_{2,4} & -y_2 & \varphi_{2,6} & \varphi_{2,7} \end{bmatrix},$$

and

$$\theta = \left[(\alpha\delta_T - \gamma)\quad \lambda\delta_T\sigma\quad r\alpha\quad \lambda r\sigma\quad \delta_T\quad \lambda\sigma\quad r\right]^T, \tag{16}$$

where $\varphi_{2,2} = -y_1^2$, $\varphi_{2,3} = -y_1 G(\lambda y_1)$, $\varphi_{2,4} = y_1^2 G(\lambda y_1)$, $\varphi_{2,6} = -y_1 y_2$, and $\varphi_{2,7} = y_2 G(\lambda y_1)$ are functions of y_1 and y_2. For presentation purposes, the arguments of A, β, and φ are suppressed in the rest of the document on the understanding that all of them are a function of y.

In order to achieve the estimation of the state variable T and unknown parameters of the model described by equations (1)-(2), we now propose the implementation of an adaptive observer for state and parameter estimation (as in [14]) for the transformed model (14) rewritten in its affine representation defined in equation (15). Such adaptive observer is described by

$$\begin{aligned} \dot{\hat{x}} &= A\hat{x} + \beta + \varphi\hat{\theta} + \{S^{-1}C^T \\ &\quad + \Lambda\Gamma^{-1}\Lambda^T C^T\} Q(y - C\hat{x}) \\ \dot{S} &= -\rho_x S - A^T S - SA + C^T QC \\ \dot{\Lambda} &= \left\{A - S^{-1}C^T QC\right\}\Lambda + \varphi \\ \dot{\Gamma} &= -\rho_\theta \Gamma + \Lambda^T C^T QC\Lambda \\ \dot{\hat{\theta}} &= \Gamma^{-1}\Lambda^T C^T Q\left(y - C\hat{x}\right), \end{aligned} \tag{17}$$

where $\hat{x} \in \Re^n$ and $\hat{\theta} \in \Re^q$ are estimations of vectors x and θ, respectively. Parameters $\rho_x \in \Re$ and $\rho_\theta \in \Re$ are sufficiently large positive constants for convergence tuning; $Q \in \Re^{d\times d}$ is a positive definite symmetric matrix; and $S \in \Re^{n\times n}$, $\Lambda \in \Re^{n\times q}$, and $\Gamma \in \Re^{q\times q}$ are solutions of $\dot{S}$, $\dot{\Lambda}$, and $\dot{\Gamma}$ in (17), respectively.

References [13], [15] provide information regarding the demonstration of exponential convergence for states and parameters using the adaptive observer (17). It is particularly important to consider the following assumptions about the system (15):

Assumption A1 [16]. Consider the state-affine system defined as

$$\begin{aligned} \dot{x} &= A(y,u)x \\ y &= Cx \end{aligned}$$

for which $\Psi(t, t_0, x_0, u)$ is its transition matrix (for the sake of simplicity, it is hereafter symbolized by Ψ), t_0 the initial time, x_0 any initial condition, and u the input signal. Assume that there exist some constant numbers $\alpha_2 > \alpha_1 > 0$ and $T_1 > 0$, such that the following condition is satisfied:

$$\alpha_1 I \le \int_t^{t+T_1} \Psi^T C^T Q(s) C\Psi ds \le \alpha_2 I,$$

for all $t \ge t_0$ and all $[t_0, x_0, u] \in \mathcal{D}$, I (the identity matrix) and $Q(s)$ are some positive definite bounded matrices of appropriate dimension.

Assumption A2 [13]. There exists a bounded time-varying matrix given by $L(t) = S^{-1}C^T Q$ such that the system $\dot{\Lambda} = (A(y,u) - L(t)C)\,\Lambda$ is exponentially stable.

Assumption A3 [17]. Assume that the function φ exhibits persistent excitation and consider the matrix Λ as the solution of $\dot{\Lambda} = \left(A - S^{-1}C^T QC\right)\Lambda + \varphi$. If there exist constants

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$\gamma_2 > \gamma_1 > 0$, $T_2 > 0$ and positive definite matrices S and Q, then the next inequality is satisfied for $t \geq t_0$:

$$\gamma_1 I \leq \int_t^{t+T_2} \Lambda^T(s)\, C^T Q(s) C \Lambda(s) ds \leq \gamma_2 I.$$

If the above assumptions are satisfied, the observation scheme given by equation (17) can be implemented to estimate the parameters θ of the system (15) under the assumption that states $y_1 := x_1 = V/\lambda$ and $y_2 := x_2 = \dot{V}/\lambda$ are measurable. Thus, solving for (13) and (16), $\hat{\alpha}$, $\hat{\sigma}$, $\hat{\gamma}$, $\hat{r}$, $\hat{\delta}_T$, and $\hat{T}$ are estimated values of the parameters $\alpha = (p-c)$, $\sigma = p/K$, $\gamma = c_T S_T$, r, δ_T and state T, respectively. In addition, assuming a case where the initial condition T_0 is known, and since $S_T = \delta_T T_0$, it is possible to compute $\hat{c}_T = \hat{\gamma}/(\hat{\delta}_T T_0)$.

IV. Numerical Results

To show the applicability of our observer scheme in SARS-CoV-2, we consider that parameter values were adjusted using data from patients with COVID-19 [8]. Note that in [8], different contagion times were used. This is attributed to the fact that the day of infection is unknown.

For numerical examples, we consider an initial simulation time as $t_0 = -3$ days. It is negative, as it refers to the period where the host does not show symptoms and, as a consequence, no measurements are taken. The initial day of observation post symptoms is t_e. The final simulation time is $t_f = 21$ days. It should be noted that viral load cannot be measured below detectable levels of about 50 copies/ml for many viral infections [8]. Implementing personalized therapies in real time may be challenging due to the need to address a number of problems, including those related to sampling and noise in measurements.

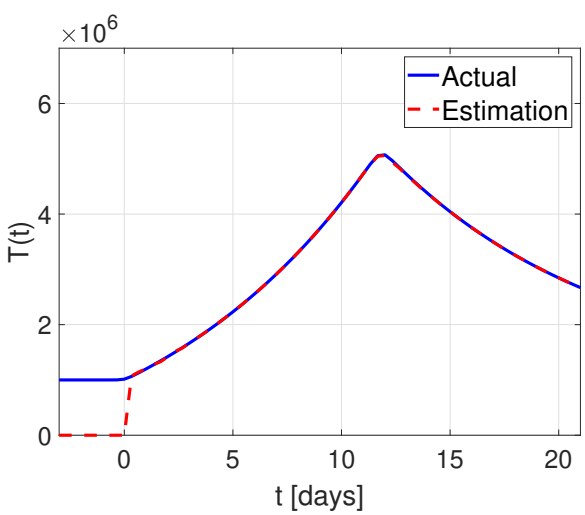

Fig. 2. State estimation of CD8+ T cell levels for SARS-CoV-2 dynamics, assuming all parameters are known.

Assuming all model parameters are known, Figure 2 shows that the observer is capable of estimating CD8+ T cells as soon as the initial measurement of viral load is recorded at $t_e = 0$. While these results are very promising, model parameters in (1)-(2) were estimated for a single patient. However, the inter-patient variability is very large, so model parameters would need to be re-calibrated. A way to tackle this problem is by including parameter estimation in the observer scheme. To this end, we consider different scenarios depending on the number of parameters to be estimated in combination with CD8+ T cells.

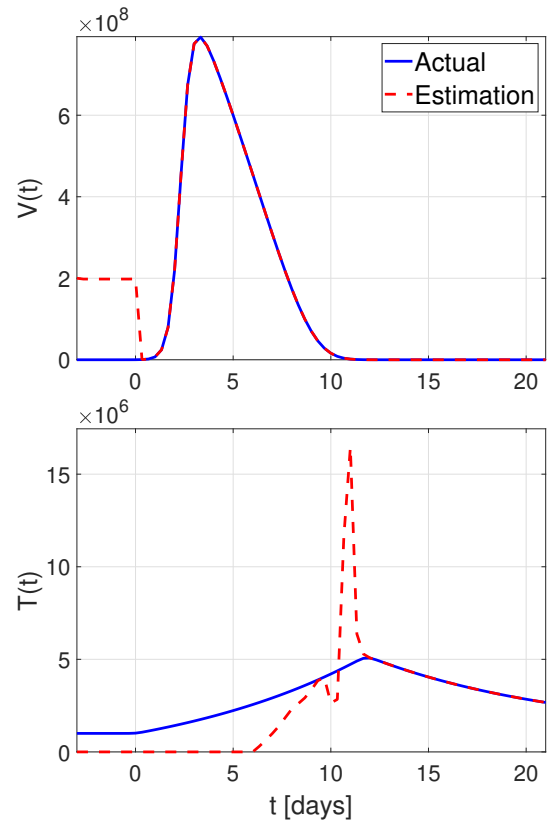

Fig. 3. State estimation for SARS-CoV-2 dynamics in Case Study 7, Table I (all parameters are unknown). Actual and estimated values for Virus and T-cell dynamics are shown in the upper and lower panels, respectively.

For quantitative comparisons, we consider the root mean square logarithmic error of the estimation $\hat{T}$ (RMSLE), which was computed as

$$RMSLE = \sqrt{\frac{1}{N}\int_{t_e}^{t_f} \left(log(T(t)) - log(\hat{T}(t))\right)^2 dt},$$

where N is the number of samples during the time interval $t = [t_e, t_f]$. Numerical results in Table I show how the cost of introducing one more parameter increases the error of the estimation of CD8+ T cells at least in a two-fold log scale. While increasing the number of parameters continues increasing the estimation error, it is not increasing linearly with respect to the number of parameters. Table I presents the different studied scenarios.

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TABLE I

DIFFERENT SCENARIOS FOR STATE AND PARAMETER ESTIMATION OF IN-HOST SARS-COV-2 MODEL (1)-(2)

Case Study	Unknown parameters	Measurable variables	T_0	RMSLE for $\hat{T}(t)$
1	All parameters are known	Only V	Unknown	5.181
2	S_T	Only V	Unknown	5.188
3	$\alpha = (p-c), S_T$	Only V	Unknown	6.060
4	$\alpha = (p-c), \sigma = \frac{p}{K}, S_T$	Only V	Unknown	8.593
5	$\alpha = (p-c), \sigma = \frac{p}{K}, \delta_T, S_T$	V and $\frac{dV}{dt}$	Unknown	9.235
6	$\alpha = (p-c), \sigma = \frac{p}{K}, \delta_T, r, S_T$	V and $\frac{dV}{dt}$	Unknown	9.234
7	$\alpha = (p-c), \sigma = \frac{p}{K}, \delta_T, r, S_T, c_T$	V and $\frac{dV}{dt}$	Required as known	9.213
8†	c_T	V and $\frac{dV}{dt}$	Unknown	5.189

† Represents a new change of coordinates with respect to equation (13), which is not shown for sake of space.
Note that cases 5-7 require the change of coordinates as in (13). Cases 1-4 do not require any change of coordinates.

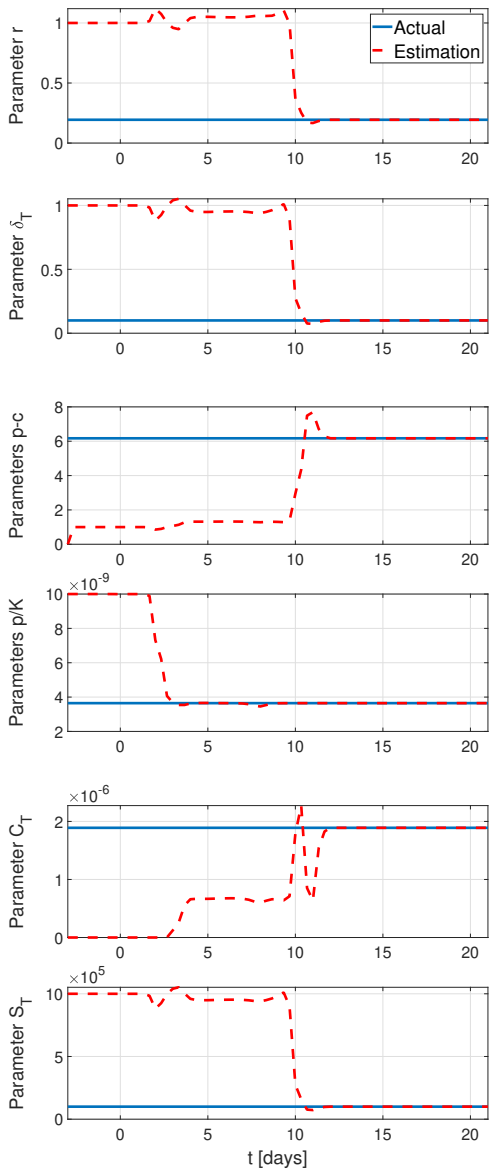

Fig. 4. Simultaneous estimation of model parameters for Case Study 7 (all parameters are unknown).

Simulations for Case Study 7, the most number of parameters estimated, are presented in Figure 3. As expected, viral load measurement is tracked accurately with the proposed strategy. Regarding CD8+ T cell dynamics, the observer required a longer period to follow T cell dynamics adequately. We need to remember the observer is not only tracking the states, but also parameter values, as we are very interested in having a methodology to tackle inter-patient variability.

Figure 4 presents the estimation of 6 model parameters simultaneously with the estimation of T cells. We can observe that most of the parameters reached the actual values between days 7 and 10 post symptoms. If we would like to improve the estimation time, we would need to reduce the number of parameters to be estimated.

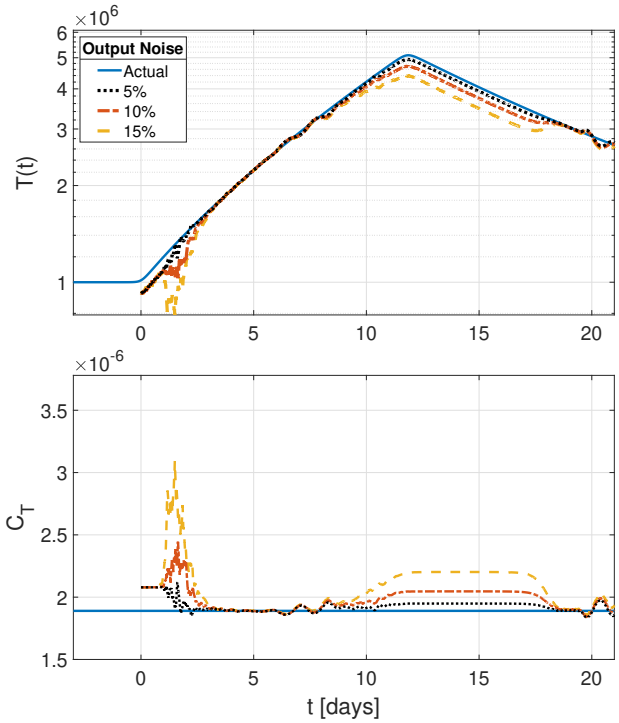

Fig. 5. Noise analysis showing adaptive observer performance when considering 5%, 10%, and 15% error in the viral load measurement. Actual and estimated values for T-cell dynamics, and C_T constant parameter are shown in the upper and lower panels, respectively.

A. Noise and Sampling analysis

To test the observer's ability to face measurement errors, output multiplicative noise is considered at different scales in the measurable viral load. The adaptive observer was

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also tested under various sampling intervals to assess its performance under data scarcity.

For noise analysis, we consider output multiplicative noise in the measured viral load, and errors of 5%, 10%, and 15% were simulated. More precisely, $\hat{V} = V_{actual} * \eta_V$, with $\eta_V \sim \mathcal{N}(1, \sigma_V)$, and σ_V taking the values of 0.05, 0.10, and 0.15. The results are presented in Figure 5. We can observe that the adaptive observer can handle multiplicative noise and still achieve good results; however, for higher errors, the performance starts to decrease.

Sampling analysis was performed to consider scarcity in data measurements. In this scenario, we consider the sampling time as 6, 12, and 24 hours, which means that we have 4, 2, and 1 viral load samples per day, respectively. The results presented in Figure 6 show that the adaptive observer can handle the estimation of the T-cell signal with one sample per day. Beyond that, the observer loses its ability to accurately estimate T-cell levels.

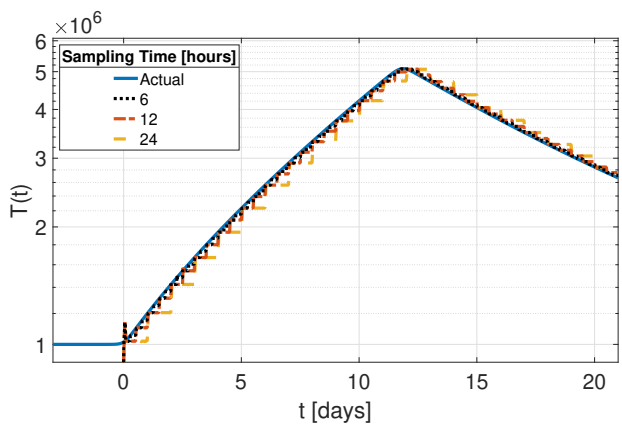

Fig. 6. Sampling analysis. Adaptive observer results when considering a sampling time of 6, 12, and 24 hours. This means 4, 2, and 1 sample of viral load per day, respectively.

As a final remark, it is important to notice that Case studies 5-8 provide another realistic implementation challenge since the algorithm relies on the derivative of the viral load, which is not directly measurable in clinical practice. Robust numerical differentiation methods, such as those in [18], offer a practical way to approximate these derivatives from biomedical sampled data with limited sampling. With appropriate windowing, these techniques enable reliable estimation of the derivative of the viral load, supporting the observer's application in clinical practice.

V. Conclusions

Tracking immune cells and estimating model parameters in respiratory infections, such as in COVID-19, is essential for improving patient-specific treatment strategies. Our analysis of observability and identifiability lays the foundation for an observer-based estimation framework that can address inter-patient variability. Through various simulation scenarios, we show the feasibility of estimating key parameters, providing critical insights for refining predictive models and enhancing clinical decision-making.

Future work should address key technical challenges, including handling noisy or incomplete experimental data, ensuring model stability during frequent updates, and developing efficient discrete-time observers that can integrate new information without excessive computational cost. Without overcoming these issues, models will remain static representations rather than adaptive digital twins that can reflect evolving biological or clinical states [19].

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